PHP Debugging

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Introduction

The article here also goes into some depth on php debugging: Setup to work with PHP code (the classic UI and the REST API)

This article sketches the major config files that process a typical http request for a php file.

Diagnosis and Mitigation

enabling devlogs

via GUI

use this clickpath

- EM7 > System > Tools > PHP Developer Logs

to enable logging and then tail these logs:

```
tail -F /data/logs/php-error.log
tail -F /data/logs/em7php.log
tail -F /data/logs/mysqli.log

```

webconfig uses a different error log for its php-fpm pool

```
 tail -F /data/logs/webconfig-php-error.log
```

via command line tool: /opt/em7/backend/silo-php/php_devlog_config.php

This file is supposed to allow you to enable devlogs, but in current versions of em7 running it produces:

```
pwd
/opt/em7/backend/silo_php

./php_devlog_config.php
Invalid silo.conf. Make sure silo.conf has a [LOCAL] section with a
valid rootdir
```

In this slack thread: https://sciencelogic.slack.com/archives/C04L3KNQLGJ/p1687354682298399 Connect your Slack account

a modified version was produced:

php_devlog_config.php
10 Apr 2024, 11:22 PM

that seems to work. This section below has more of the background story here: PHP Debugging

NOTE: both the modified and broken versions of this script above are problematic vs the original version. The original forced the user to use sudo -u s-em7-http to run the script and would fail as any other user. this ensured that the CLI and WEB both accessed the config as the same user, or at least the same user group. Later versions of the script tried to “improve” this design and introduced breakage either for CLI logging, UI logging, or both due to file permissions on the cached config.
12.5 and up have a working tool again.

pre 8.12.1 edit of /etc/silo.conf

Adding a section like this to /etc/silo.conf:

```
        [DEVLOG]
        filefunclen = 54
        logfile = /data/logs/em7_php_dev.log
        logformat = "%s|%-54s|%-4s|%s|%s"
        loglevel = 63
        logvars = time,filefunc,line,level,msg
        mysqli = /data/logs/mysqli_queries.log
        timeformat = H:i:

```

used to have the effect of enabling devlogs pre-8.12.1.

nginx access and error logs

Are configured with these lines in /etc/nginx/conf.d/em7ngx_web_ui.conf

```
    access_log
syslog:server=unix:/var/run/rsyslog.sock,facility=local7,tag=em7silo_ng
inx,severity=info,nohostname;
    error_log
syslog:server=unix:/var/run/rsyslog.sock,tag=em7silo_nginx,nohostname
info;
```

And in oracle linux syslogs are written to /var/log/messages. This command line would allow you to see error and access logs:

```
tail -F /var/log/messages | grep em7silo_nginx
```

Here is example output from accessing my test CDB with http://10.2.4.50

```
2023-10-12T16:11:32.566499+00:00 cdb em7silo_nginx[1259]: 10.2.4.56 - - [12/Oct/2023:16:11:32 +0000] "GET / HTTP/2.0" 302 0 "-" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/118.0"
2023-10-12T16:11:32.576450+00:00 cdb em7silo_nginx[1259]: 10.2.4.56 - - [12/Oct/2023:16:11:32 +0000] "GET /ap2 HTTP/2.0" 200 166 "-" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/118.0"
2023-10-12T16:11:32.817119+00:00 cdb em7silo_nginx[1259]: 2023/10/12 16:11:32 [error] 1259#1259: *12887 connect() failed (111: Connection refused) while connecting to upstream, client: 10.2.4.56, server: , request: "GET /loginTheme HTTP/2.0", upstream: "http://[::1]:3000/loginTheme", host: "10.2.4.50", referrer: "https://10.2.4.50/ap2"
2023-10-12T16:11:32.817133+00:00 cdb em7silo_nginx[1259]: 2023/10/12 16:11:32 [warn] 1259#1259: *12887 upstream server temporarily disabled while connecting to upstream, client: 10.2.4.56, server: , request: "GET /loginTheme HTTP/2.0", upstream: "http://[::1]:3000/loginTheme", host: "10.2.4.50", referrer: "https://10.2.4.50/ap2"
2023-10-12T16:11:32.820300+00:00 cdb em7silo_nginx[1259]: 10.2.4.56 - - [12/Oct/2023:16:11:32 +0000] "GET /images/commonSprites.svg HTTP/2.0" 304 0 "https://10.2.4.50/ap2" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/118.0"
2023-10-12T16:11:32.823027+00:00 cdb em7silo_nginx[1259]: 10.2.4.56 - - [12/Oct/2023:16:11:32 +0000] "GET /loginTheme HTTP/2.0" 200 817 "https://10.2.4.50/ap2" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/118.0"
2023-10-12T16:11:33.408193+00:00 cdb em7silo_nginx[1259]: 10.2.4.56 - - [12/Oct/2023:16:11:33 +0000] "GET /authenticate/check HTTP/2.0" 200 63 "https://10.2.4.50/ap2" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/118.0"
2023-10-12T16:11:33.446077+00:00 cdb em7silo_nginx[1259]: 10.2.4.56 - - [12/Oct/2023:16:11:33 +0000] "GET /loginTheme HTTP/2.0" 200 817 "https://10.2.4.50/ap2" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/118.0"
2023-10-12T16:11:33.545019+00:00 cdb em7silo_nginx[1259]: 10.2.4.56 - - [12/Oct/2023:16:11:33 +0000] "GET /loginTheme HTTP/2.0" 200 817 "https://10.2.4.50/ap2" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/118.0"

```

On an 11.3.2 system we see these lines in /etc/nginx/conf.d/em7ngx_web_ui.conf:

```
    access_log  /var/log/em7/em7ngx.access.log  main;
    error_log   /var/log/em7/ngx.log            info;
```

and so we would use this to view nginx accesses:

```
tail -F /var/log/em7/em7ngx.access.log
```

php-fpm error and access logging

You wouldn't normally want to enable access logging for php-fpm, but it could make sense when debugging behavior on a test stack.

One way to do this would be to uncomment this stanza:

```
;; The access log file
;; Default: not set
;;access.log = log/$pool.access.log
access.log = /var/log/$pool.access.log

```

and then bounce php-fpm with:

```
systemctl restart php-fpm
```

There are two pools defined on em7:

- em7 (via /etc/php-fpm.d/em7.conf) and
- em7-webconfig (via /etc/php-fpm.d/em7-webconfig.conf)

So you would expect them to log here, given the above config

- em7: /var/log/em7.access.log
- em7-webconfig: /var/log/em7-webconfig.access.log

Using backtraces and mysqli log to uncover theory of operation: ticket template registry page

As with all large bodies of source code in all languages, seemingly simple GUI operations (like navigating to registry → ticketting → templates and viewing the list of ticket templates) involve vast numbers of far flung segments of code cooperating to achieve them. Attempting to gain an understanding of how files, classes, functions, and critical lines of code relate to that operation (the theory of operation) by only reading the source code is often a fool's errand. However, by instrumenting the front end code and then exercising the feature in question, it is possible to gain lots of insight quickly, and to document what you have found.

This section shows how to use

- php's debug_backtrace() and
- em7's mysqli logging

to make uncovering the theory of operation

- relatively quick
- easy to write down so you can review it later / share with others

The documentation system I use for this is written up here: A Proposed Project Documentation Practice: FIXME Commentary . You don't need to read the whole thing right now but reading the intro section would help you follow the rest of this section.

A Map of the Theory of Operation

In this section, we show what documenting the theory of operation looks like. Further down, we will get into how to efficiently gather the information in the first place.

If you are curious, I have gathered the feature branch I added this commentary into this git bundle: bundle

You can pull this into your git repo with these commands:

```
git remote add some_bundle bundle
git ls-remote some_bundle
dfde4b618b10b1c8b3292c65ac3027865e85094f refs/heads/research_classic_ui
git fetch some_bundle research_classic_ui:research_classic_ui
From /home/chris/tmp/bundle
 * [new branch]            research_classic_ui -> research_classic_ui
git checkout research_classic_ui
```

structural overview

The first thing you want to know, if nothing else, is what database tables and source code classes and functions are involved. This

```
./sql/schemas/em7_db/master_biz/ticket_templates.sql:15:/* FIXME_INFO_TYPE: ticket_templates {*/
./sql/schemas/em7_db/master_biz/ticket_templates.sql:39:/* FIXME_INFO_TYPE: } ticket_templates*/
./gui/ap/classes/db_registry_table.inc:4:// FIXME_INFO_CLASS: db_registry_table extends registry_table {
./gui/ap/classes/db_registry_table.inc:397:// FIXME_INFO_FN: fetch_data() {
./gui/ap/classes/db_registry_table.inc:517:// FIXME_INFO_FN: } fetch_data()
./gui/ap/classes/db_registry_table.inc:561:// FIXME_INFO_FN: get_dynamic_array() {
./gui/ap/classes/db_registry_table.inc:570:// FIXME_INFO_FN: } get_dynamic_array()
./gui/ap/classes/db_registry_table.inc:688:// FIXME_INFO_CLASS: } db_registry_table extends registry_table
./gui/ap/classes/registry_table.inc:4:// FIXME_INFO_CLASS: registry_table {
./gui/ap/classes/registry_table.inc:570:// FIXME_INFO_FN: echo_init_script() {
./gui/ap/classes/registry_table.inc:579:// FIXME_INFO_FN: } echo_init_script()
./gui/ap/classes/registry_table.inc:855:// FIXME_INFO_FN: get_dynamic_json() {
./gui/ap/classes/registry_table.inc:860:// FIXME_INFO_FN: } get_dynamic_json()
./gui/ap/classes/registry_table.inc:1389:// FIXME_INFO_CLASS: } registry_table
./gui/ap/includes/functions_core.inc:1318:// FIXME_INFO_FN: page_include() {
./gui/ap/includes/functions_core.inc:1344:// FIXME_INFO_FN: } page_include()
./gui/ap/includes/functions_core.inc:1358:// FIXME_INFO_FN: phtml_include() {
./gui/ap/includes/functions_core.inc:1366:// FIXME_INFO_FN: } phtml_include()

```

shows us that

- master_biz.ticket_templates is the main database table and that
- the primary classes and methods are
  - db_registry_table
    - fetch_data()
    - get_dynamic_array()
  - registry_table
    - echo_init_script()
    - get_dynamic_json()
- while these functions are also important
  - page_include()
  - phtml_include()

This summary was constructed by placing various comments, directly in the files of a git working tree of the php code (in the central em7 repo, on a feature branch):

- FIXME_INFO_CLASS: to mark the beginning and end of the code segments defining a whole class
- FIXME_INFO_FN: to mark the beginning and end of methods and top level functions and
- FIXME_INFO_TYPE: to mark the beginning and end of database tables

and then running this command line

```
fixme_report.sh | fixme_struct.sh
```

involving these scripts: A Proposed Project Documentation Practice: FIXME Commentary | AProposedProjectDocumentationPractice:FIXMECommentary AProposedProjectDocumenta...

and only then, manually editing the results. The key to making this style of documentation work is to drop comments of various types directly into the source code and then running various combinations of find+grep to extract meaningful reports. This allows it to proceed as a natural extension of reading and tinkering with code, instead of as a jarring interruption of it.

telling the story (asking and answering questions, FIXME_INFO_ISSUE_X_Y)

The structural overview above came from running this larger map through fixme_struct.sh

```
./project_notes/notes.txt:2:# FIXME_INFO_FILE:
./project_notes/notes.txt:35:# FIXME_INFO_ISSUE_3_2: we load the URL below (http://10.2.4.50/em7/index.em7?exec=registry&act=registry_ticket_jobs&ap2_embedded=1)
./project_notes/notes.txt:48:# FIXME_INFO_ISSUE_3_3: and we enable PHP debug logging via em7 -> system -> tools -> php developer logs
./project_notes/notes.txt:49:# FIXME_INFO_ISSUE_3_4: and we leave this running on the CDB: tail -F /data/logs/mysqli.log > run1.txt
./project_notes/notes.txt:50:# FIXME_INFO_ISSUE_3_5: which produces these two queries
./project_notes/notes.txt:73:# FIXME_INFO_ISSUE_3_9: the backtrace recorded in the devlog is below.  Following this up the call stack takes us

./sql/schemas/em7_db/master_biz/ticket_templates.sql:15:/* FIXME_INFO_TYPE: ticket_templates {*/
./sql/schemas/em7_db/master_biz/ticket_templates.sql:17:/* FIXME_INFO_ISSUE_3_1: how does the php code that displays the ticket template registry wind up querying this table */
./sql/schemas/em7_db/master_biz/ticket_templates.sql:39:/* FIXME_INFO_TYPE: } ticket_templates */

./phpshare/classes/system/devlog.inc:3:// FIXME_INFO_CLASS: devlog {
./phpshare/classes/system/devlog.inc:87:// FIXME_INFO_FN: fixmelog() {
./phpshare/classes/system/devlog.inc:89:// FIXME_INFO_ISSUE_3_7: this function, and then calling it from
./phpshare/classes/system/devlog.inc:96:// FIXME_INFO_FN: } fixmelog()
./phpshare/classes/system/devlog.inc:663:// FIXME_INFO_CLASS: } devlog

./gui/ap/classes/db_registry_table.inc:4:// FIXME_INFO_CLASS: db_registry_table extends registry_table {
./gui/ap/classes/db_registry_table.inc:397:// FIXME_INFO_FN: fetch_data() {
./gui/ap/classes/db_registry_table.inc:399:// FIXME_INFO_ISSUE_3_8: here, and following the traceback.
./gui/ap/classes/db_registry_table.inc:427:// FIXME_INFO_ISSUE_3_6: which takes us here.  We can learn more about this execution context by gathering a bracktrace by writing
./gui/ap/classes/db_registry_table.inc:517:// FIXME_INFO_FN: } fetch_data()
./gui/ap/classes/db_registry_table.inc:561:// FIXME_INFO_FN: get_dynamic_array() {
./gui/ap/classes/db_registry_table.inc:565:// FIXME_INFO_ISSUE_3_10:
./gui/ap/classes/db_registry_table.inc:570:// FIXME_INFO_FN: } get_dynamic_array()
./gui/ap/classes/db_registry_table.inc:688:// FIXME_INFO_CLASS: } db_registry_table extends registry_table

./gui/ap/classes/registry_table.inc:4:// FIXME_INFO_CLASS: registry_table {
./gui/ap/classes/registry_table.inc:570:// FIXME_INFO_FN: echo_init_script() {
./gui/ap/classes/registry_table.inc:573:// FIXME_INFO_ISSUE_3_12:
./gui/ap/classes/registry_table.inc:579:// FIXME_INFO_FN: } echo_init_script()
./gui/ap/classes/registry_table.inc:855:// FIXME_INFO_FN: get_dynamic_json() {
./gui/ap/classes/registry_table.inc:857:// FIXME_INFO_ISSUE_3_11:
./gui/ap/classes/registry_table.inc:860:// FIXME_INFO_FN: } get_dynamic_json()
./gui/ap/classes/registry_table.inc:1389:// FIXME_INFO_CLASS: } registry_table

./gui/ap/phtml/common_regtable.phtml:3:<? // FIXME_INFO_ISSUE_3_14: ?>

./gui/ap/phtml/common_regtable_liquid.phtml:38:<? // FIXME_INFO_ISSUE_3_13: ?>

./gui/ap/includes/functions_core.inc:1318:// FIXME_INFO_FN: page_include() {
./gui/ap/includes/functions_core.inc:1331:// FIXME_INFO_ISSUE_3_15:
./gui/ap/includes/functions_core.inc:1344:// FIXME_INFO_FN: } page_include()
./gui/ap/includes/functions_core.inc:1358:// FIXME_INFO_FN: phtml_include() {
./gui/ap/includes/functions_core.inc:1363:// FIXME_INFO_ISSUE_3_16:
./gui/ap/includes/functions_core.inc:1366:// FIXME_INFO_FN: } phtml_include()

./gui/ap/exec/common_regtable.inc:79:// FIXME_INFO_ISSUE_3_17:
```

```

./gui/ap/includes/liquid_exec_common.inc:37:// FIXME_INFO_ISSUE_3_18:

./gui/ap/www/em7/index.em7:162:// FIXME_INFO_ISSUE_3_19:

```

The additional comments scattered throughout (FIXME_INFO_ISSUE_3_1 ... FIXME_INFO_ISSUE_3_19), when gathered together in series, look like this:

```
fixme_report.sh | fixme_issue.py 3
./sql/schemas/em7_db/master_biz/ticket_templates.sql:17:/* FIXME_INFO_ISSUE_3_1: how does the php code that displays the ticket template registry wind up querying this table */
./project_notes/notes.txt:35:# FIXME_INFO_ISSUE_3_2: we load the URL below (http://10.2.4.50/em7/index.em7?exec=registry&act=registry_ticket_jobs&ap2_embedded=1)
./project_notes/notes.txt:48:# FIXME_INFO_ISSUE_3_3: and we enable PHP debug logging via em7 -> system -> tools -> php developer logs
./project_notes/notes.txt:49:# FIXME_INFO_ISSUE_3_4: and we leave this running on the CDB: tail -F /data/logs/mysqli.log > run1.txt
./project_notes/notes.txt:50:# FIXME_INFO_ISSUE_3_5: which produces these two queries
./gui/ap/classes/db_registry_table.inc:427:// FIXME_INFO_ISSUE_3_6: which takes us here.  We can learn more about this execution context by gathering a bracktrace by writing
./phpshare/classes/system/devlog.inc:89:// FIXME_INFO_ISSUE_3_7: this function, and then calling it from
./gui/ap/classes/db_registry_table.inc:399:// FIXME_INFO_ISSUE_3_8: here, and following the traceback.
./project_notes/notes.txt:73:# FIXME_INFO_ISSUE_3_9: the backtrace recorded in the devlog is below.  Following this up the call stack takes us
./gui/ap/classes/db_registry_table.inc:565:// FIXME_INFO_ISSUE_3_10:
./gui/ap/classes/registry_table.inc:857:// FIXME_INFO_ISSUE_3_11:
./gui/ap/classes/registry_table.inc:573:// FIXME_INFO_ISSUE_3_12:
./gui/ap/phtml/common_regtable_liquid.phtml:38:<? // FIXME_INFO_ISSUE_3_13: ?>
./gui/ap/phtml/common_regtable.phtml:3:<? // FIXME_INFO_ISSUE_3_14: ?>
./gui/ap/includes/functions_core.inc:1331:// FIXME_INFO_ISSUE_3_15:
./gui/ap/includes/functions_core.inc:1363:// FIXME_INFO_ISSUE_3_16:
./gui/ap/exec/common_regtable.inc:79:// FIXME_INFO_ISSUE_3_17:
./gui/ap/includes/liquid_exec_common.inc:37:// FIXME_INFO_ISSUE_3_18:
./gui/ap/www/em7/index.em7:162:// FIXME_INFO_ISSUE_3_19:

```

Note that

- the series begins by asking the question (and then answering): how does the php code that displays the ticket template registry wind up querying this table?
- where ‘this table’ is master_biz.ticket_templates (revealed in the filename the comment was written in)
- comments 3_2 ... 3_6 describe how we gathered a php mysqli log, identifying the line of code that produced it
- 3_7 ... 3_9 show how we gathered this php traceback:

```
16:59:38.65827|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |90  |error|    [0] => "db_registry_table.fetch(): enter"
16:59:38.65832|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |90  |error|    [1] => "include(string) at /opt/em7/gui/ap/www/em7/index.em7:162"
16:59:38.65838|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |90  |error|    [2] => "require(string) at /opt/em7/gui/ap/includes/liquid_exec_common.inc:37"
16:59:38.65842|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |90  |error|    [3] => "phtml_include(string,array) at /opt/em7/gui/ap/exec/common_regtable.inc:79"
16:59:38.65847|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |90  |error|    [4] => "page_include(string,array) at /opt/em7/gui/ap/includes/functions_core.inc:1359"
16:59:38.65851|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |90  |error|    [5] => "include(string) at /opt/em7/gui/ap/includes/functions_core.inc:1330"
16:59:38.65856|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |90  |error|    [6] => "require(string) at /opt/em7/gui/ap/phtml/common_regtable.phtml:2"
16:59:38.65861|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |90  |error|    [7] => "echo_init_script() at /opt/em7/gui/ap/phtml/common_regtable_liquid.phtml:37"
16:59:38.65866|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |90  |error|    [8] => "get_dynamic_json() at /opt/em7/gui/ap/classes/registry_table.inc:571"
16:59:38.65871|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |90  |error|    [9] => "get_dynamic_array() at /opt/em7/gui/ap/classes/registry_table.inc:852"
16:59:38.65875|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |90  |error|    [10] => "fetch_data() at /opt/em7/gui/ap/classes/db_registry_table.inc:563"
16:59:38.65880|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |90  |error|    [11] => "fixmelog(string,boolean) at /opt/em7/gui/ap/classes/db_registry_table.inc:398"
```

- 3_10 ... through 3_19 then record the location of each line of code in the traceback by being placed before them in the code

To arrive at the report quoted at the top of this section, we did some manual editing of the output of

```
fixme_report.sh | fixme_struct.sh 3
```

From that report, it is easy to see the context around each point in the traceback. 3_8 (where the query is run) occurred in class.method() = db_registry_table.fetch_data(), in the PHP source file G3code/gui/ap/classes/db_registry_table.inc.

That report functions as a map of where important things are happening, and where more detail can be found. Looking inside of G3code/project_notes/notes.txt, for example, we can see the full text of the queries we captured from the mysqli log:

```
# FIXME_INFO_ISSUE_3_5: which produces these two queries
fetch_data()/opt/em7/gui/ap/classes/db_registry_table.inc [line:407  calling:autofetch_value]
SELECT  COUNT(*) FROM master_biz.ticket_templates ts
 LEFT JOIN master_access.accounts a ON (ts.edit_user=a.uid)
 LEFT JOIN master_biz.user_information a_info ON (a_info.uid = a.uid)
 LEFT JOIN master.auth_userstore_attribute_map auth_resource ON (auth_resource.resource_id = a.resource_id)
WHERE 1 LIMIT 1
SIZE: 306 bytes
ROWS:1|ERRNO:0|MSG:
DURATION: 0.00036 seconds

fetch_data()/opt/em7/gui/ap/classes/db_registry_table.inc [line:423  calling:autofetch_all]
SELECT ts.name, ts.sch_id, ts.ticket_descr, ts.ticket_severity, IF(auth_resource.userdisplayname="disable" OR a_info.userdisplayname IS NULL OR a_info.userdisplayname="", a.user, a_info.userdisplayname) AS user, UNIX_TIMESTAMP(ts.edit_date) AS edit_date, (template_feature = 2 AND (SELECT v.removed FROM scheduler.vevent v WHERE v.removed = 0 AND (v.dtstart > NOW() OR v.recur_until > NOW()) LIMIT 1)) AS sch_pending, (template_feature = 1 AND (SELECT ticket_template FROM master.policies_actions_ticket WHERE ticket_template = ts.sch_id LIMIT 1)) AS automated, ts.ppguid
FROM master_biz.ticket_templates ts
 LEFT JOIN master_access.accounts a ON (ts.edit_user=a.uid)
 LEFT JOIN master_biz.user_information a_info ON (a_info.uid = a.uid)
 LEFT JOIN master.auth_userstore_attribute_map auth_resource ON (auth_resource.resource_id = a.resource_id)
WHERE 1
ORDER BY name ASC LIMIT 50
SIZE: 880 bytes
ROWS:1|ERRNO:0|MSG:
DURATION: 0.00058 seconds

```

Disentangling the New and Classic UI interactions

Once we navigate to registry → ticketing → templates, we:

- leave this command line running on the CDB (this will dump the resulting URL accesses as nginx sees them):

```
tail --lines=0  -F /var/log/messages | grep em7silo_nginx > run1.txt
```

- refresh the page
- kill the tail command

The resulting list of accesses:

```
2023-10-12T22:19:24.074875+00:00 cdb em7silo_nginx[1259]: 10.2.4.56 - - [12/Oct/2023:22:19:24 +0000] "GET /frame/TicketTemplates HTTP/2.0" 200 166 "-" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/118.0"
2023-10-12T22:19:24.402489+00:00 cdb em7silo_nginx[1259]: 10.2.4.56 - - [12/Oct/2023:22:19:24 +0000] "GET /fonts/lato-normal.75614cfcfedd509b1f7a.woff2 HTTP/2.0" 499 0 "https://10.2.4.50/frame/TicketTemplates" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/118.0"
2023-10-12T22:19:24.410495+00:00 cdb em7silo_nginx[1259]: 10.2.4.56 - - [12/Oct/2023:22:19:24 +0000] "GET /images/commonSprites.svg HTTP/2.0" 304 0 "https://10.2.4.50/frame/TicketTemplates" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/118.0"
2023-10-12T22:19:24.432622+00:00 cdb em7silo_nginx[1259]: 10.2.4.56 - - [12/Oct/2023:22:19:24 +0000] "GET /loginTheme HTTP/2.0" 200 817 "https://10.2.4.50/frame/TicketTemplates" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/118.0"
2023-10-12T22:19:24.443660+00:00 cdb em7silo_nginx[1259]: 2023/10/12 22:19:24 [error] 1259#1259: *13494 open() "/usr/local/silo/gui/ap/www/favicon.ico" failed (2: No such file or directory), client: 10.2.4.56, server: , request: "GET /favicon.ico HTTP/2.0", host: "10.2.4.50", referrer: "https://10.2.4.50/frame/TicketTemplates"
2023-10-12T22:19:24.443734+00:00 cdb em7silo_nginx[1259]: 10.2.4.56 - - [12/Oct/2023:22:19:24 +0000] "GET /favicon.ico HTTP/2.0" 404 146 "https://10.2.4.50/frame/TicketTemplates" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/118.0"
2023-10-12T22:19:25.311285+00:00 cdb em7silo_nginx[1257]: 127.0.0.1 - - [12/Oct/2023:22:19:25 +0000] "GET /api/account/_self HTTP/1.1" 200 2083 "-" "-"
2023-10-12T22:19:25.360418+00:00 cdb em7silo_nginx[1259]: 127.0.0.1 - - [12/Oct/2023:22:19:25 +0000] "GET /api/account/_self HTTP/1.1" 200 2083 "-" "-"
2023-10-12T22:19:25.360939+00:00 cdb em7silo_nginx[1261]: 127.0.0.1 - - [12/Oct/2023:22:19:25 +0000] "GET /api/account/_self/access_hooks HTTP/1.1" 200 11678 "-" "-"
2023-10-12T22:19:25.369517+00:00 cdb em7silo_nginx[1257]: 127.0.0.1 - - [12/Oct/2023:22:19:25 +0000] "GET /api/account/_self/access_hooks HTTP/1.1" 200 11671 "-" "-"
2023-10-12T22:19:25.422029+00:00 cdb em7silo_nginx[1257]: 127.0.0.1 - - [12/Oct/2023:22:19:25 +0000] "GET /api/organization?hide_filterinfo=1&limit=100000&order._id=asc HTTP/1.1" 200 127 "-" "-"
2023-10-12T22:19:25.422235+00:00 cdb em7silo_nginx[1261]: 127.0.0.1 - - [12/Oct/2023:22:19:25 +0000] "GET /api/organization?hide_filterinfo=1&limit=100000&order._id=asc HTTP/1.1" 200 127 "-" "-"
2023-10-12T22:19:25.534059+00:00 cdb em7silo_nginx[1257]: 127.0.0.1 - - [12/Oct/2023:22:19:25 +0000] "GET /api/definition/system_settings HTTP/1.1" 200 5547 "-" "-"
2023-10-12T22:19:25.537214+00:00 cdb em7silo_nginx[1259]: 10.2.4.56 - - [12/Oct/2023:22:19:25 +0000] "POST /gql?operation=SystemSettingsAndReleaseInfo HTTP/2.0" 200 650 "https://10.2.4.50/frame/TicketTemplates" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/118.0"
2023-10-12T22:19:25.571742+00:00 cdb em7silo_nginx[1257]: 127.0.0.1 - - [12/Oct/2023:22:19:25 +0000] "GET /api/account/_self HTTP/1.1" 200 2083 "-" "-"
2023-10-12T22:19:25.640031+00:00 cdb em7silo_nginx[1257]: 127.0.0.1 - - [12/Oct/2023:22:19:25 +0000] "GET /api/account/1/access_hooks HTTP/1.1" 200 11678 "-" "-"
```

```
2023-10-12T22:19:25.646698+00:00 cdb em7silo_nginx[1259]: 10.2.4.56 - - [12/Oct/2023:22:19:25 +0000] "POST /gql?operation=getUserInfoWithTogglesAndSystemVariables HTTP/2.0" 200 22641 "https://10.2.4.50/frame/TicketTemplates" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/118.0"
2023-10-12T22:19:26.061576+00:00 cdb em7silo_nginx[1259]: 10.2.4.56 - - [12/Oct/2023:22:19:26 +0000] "POST /gql?operation=featureToggles HTTP/2.0" 200 2547 "https://10.2.4.50/frame/TicketTemplates" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/118.0"
2023-10-12T22:19:26.063650+00:00 cdb em7silo_nginx[1259]: 10.2.4.56 - - [12/Oct/2023:22:19:26 +0000] "POST /gql?operation=logPageAccess HTTP/2.0" 200 45 "https://10.2.4.50/frame/TicketTemplates" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/118.0"
2023-10-12T22:19:26.199742+00:00 cdb em7silo_nginx[1259]: 10.2.4.56 - - [12/Oct/2023:22:19:26 +0000] "GET /images/navSprites.svg HTTP/2.0" 304 0 "https://10.2.4.50/frame/TicketTemplates" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/118.0"
2023-10-12T22:19:26.255261+00:00 cdb em7silo_nginx[1261]: 127.0.0.1 - - [12/Oct/2023:22:19:26 +0000] "GET /api/event?extended_fetch=1&limit=10&offset=0&filter.0.aligned_resource%2Fdevice.in=1%2C2&filter.0.event_policy.in=1904%2C1867%2C1888%2C1941%2C1866%2C1896%2C1900%2C1871%2C1853%2C1891%2C1875%2C1955%2C2014%2C1858%2C1898%2C1926%2C1958%2C1886%2C1882%2C1856%2C1937%2C1849%2C1873%2C1935%2C1939%2C1884%2C1943%2C1931%2C1894&order.date_active=desc&order._id=asc HTTP/1.1" 200 2951 "-" "-"
2023-10-12T22:19:26.256076+00:00 cdb em7silo_nginx[1261]: 2023/10/12 22:19:26 [info] 1261#1261: *13526 client 127.0.0.1 closed keepalive connection
2023-10-12T22:19:26.257145+00:00 cdb em7silo_nginx[1259]: 10.2.4.56 - - [12/Oct/2023:22:19:26 +0000] "POST /gql?operation=AdminNotifierEventCount HTTP/2.0" 200 249 "https://10.2.4.50/frame/TicketTemplates" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/118.0"
2023-10-12T22:19:26.267562+00:00 cdb em7silo_nginx[1259]: 10.2.4.56 - - [12/Oct/2023:22:19:26 +0000] "GET /em7/index.em7?exec=registry&act=registry_ticket_jobs&ap2_embedded=1 HTTP/2.0" 200 4150 "https://10.2.4.50/frame/TicketTemplates" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/118.0"
2023-10-12T22:19:26.410086+00:00 cdb em7silo_nginx[1259]: 10.2.4.56 - - [12/Oct/2023:22:19:26 +0000] "GET /em7/libs/css.em7/1/registry_table?1697149166 HTTP/2.0" 200 18701 "https://10.2.4.50/em7/index.em7?exec=registry&act=registry_ticket_jobs&ap2_embedded=1" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/118.0"
2023-10-12T22:19:26.481743+00:00 cdb em7silo_nginx[1259]: 10.2.4.56 - - [12/Oct/2023:22:19:26 +0000] "GET /em7/libs/buttons/edit.png HTTP/2.0" 200 680 "https://10.2.4.50/em7/index.em7?exec=registry&act=registry_ticket_jobs&ap2_embedded=1" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/118.0"

```

consists of

- a mixture of the new UI accesses (routed to the node.js server) and
- PHP accesses that go through the fastcgi server processes (php-fpm) to load and execute php files (the classic UI)

From the above list, we look for all of the URLs starting with “/em7”, arriving at just the last two lines

```
2023-10-30T16:52:02.108532+00:00 cdb em7silo_nginx[1259]: 10.2.4.56 - - [30/Oct/2023:16:52:02 +0000] "GET /em7/index.em7?exec=registry&act=registry_ticket_jobs&ap2_embedded=1 HTTP/2.0" 200 4150 "https://10.2.4.50/frame/TicketTemplates" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/119.0"
2023-10-30T16:52:02.230401+00:00 cdb em7silo_nginx[1259]: 10.2.4.56 - - [30/Oct/2023:16:52:02 +0000] "GET /em7/libs/css.em7/1/registry_table?1698684722 HTTP/2.0" 200 18701
```

If we open a new brower tab and enter the classic UI URL (in this case, my test CDB was at 10.2.4.50):

```
https://10.2.4.50/em7/index.em7?exec=registry&act=registry_ticket_jobs&ap2_embedded=1
```

and do a refresh, while watching the output of

```
tail -F /var/log/messages | grep em7silo_nginx
```

we see these accesses being generated

```
2023-10-30T17:08:42.693517+00:00 cdb em7silo_nginx[1259]: 10.2.4.56 - - [30/Oct/2023:17:08:42 +0000] "GET /em7/index.em7?exec=registry&act=registry_ticket_jobs&ap2_embedded=1 HTTP/2.0" 200 4149 "-" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/119.0"
2023-10-30T17:08:42.760835+00:00 cdb em7silo_nginx[1259]: 10.2.4.56 - - [30/Oct/2023:17:08:42 +0000] "GET /em7/libs/css.em7/1/registry_table?1698685722 HTTP/2.0" 200 18701 "https://10.2.4.50/em7/index.em7?exec=registry&act=registry_ticket_jobs&ap2_embedded=1" "Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:109.0) Gecko/20100101 Firefox/119.0"

```

thus we conclude that the URL tied to loading the registry page is: https://10.2.4.50/em7/index.em7?exec=registry&act=registry_ticket_jobs&ap2_embedded=1

Capturing Relevant Queries, and Identifying the PHP code behind them

Now that we’ve isolated a URL that loads the page we are interested in we:

- turn on php devlogging via this click path: EM7 > System > Tools > PHP Developer Logs
- start tailing the mysqli log

```
tail -F /data/logs/mysqli.log > run1.txt
```

- reload our url to render the ticketing template page:

```
https://10.2.4.50/em7/index.em7?exec=registry&act=registry_ticket_jobs&ap2_embedded=1
```

- allow this to finish, then kill the tail, resulting in

```
wc run1.txt
  265  1491 20697 run1.txt

cat run1.txt

_get_extended_tables()/opt/em7/phpshare/classes/entity/_delegate/d_ent_custom_fields.inc [line:288  calling:autofetch_column]
SELECT CONCAT(`table_schema`,'.',`table_name`) FROM `information_schema`.`tables` WHERE `table_schema` = 'master_custom' AND `table_name` LIKE 'ent_css_custom_extended%'
SIZE: 170 bytes
ROWS:0|ERRNO:0|MSG:
DURATION: 0.00035 seconds

...

_get_extended_tables()/opt/em7/phpshare/classes/entity/_delegate/d_ent_custom_fields.inc [line:288  calling:autofetch_column]
SELECT CONCAT(`table_schema`,'.',`table_name`) FROM `information_schema`.`tables` WHERE `table_schema` = 'master_custom' AND `table_name` LIKE 'ent_css_custom_extended%'
SIZE: 170 bytes
ROWS:0|ERRNO:0|MSG:
DURATION: 0.00049 seconds

```

- searching this file just for the php lines of code that executed queries, we see this sequence of 34:

```
grep "\[" run1.txt

_get_extended_tables()/opt/em7/phpshare/classes/entity/_delegate/d_ent_custom_fields.inc [line:288  calling:autofetch_column]
__construct()/opt/em7/phpshare/classes/system/authentication_profile.inc [line:34  calling:autofetch_all]
__construct()/opt/em7/phpshare/classes/system/authentication_profile.inc [line:52  calling:autofetch_all]
__construct()/opt/em7/phpshare/classes/system/authentication_profile.inc [line:64  calling:autofetch_all]
active_userstore()/opt/em7/phpshare/classes/system/authentication_profile.inc [line:331  calling:autofetch_value]
_fetch_multi_where()/opt/em7/phpshare/classes/entity/entity_common.inc [line:398  calling:autofetch_all]
_base_fetch_join_fields__impl()/opt/em7/phpshare/classes/entity/entity_common.inc [line:1123  calling:autofetch_all]
_base_fetch_join_fields__impl()/opt/em7/phpshare/classes/entity/entity_common.inc [line:1123  calling:autofetch_all]
fetch_extra_data__impl()/opt/em7/phpshare/classes/entity/_delegate/d_ent_foreign_relations.inc [line:69  calling:query_helper]
fetch_extra_data__impl()/opt/em7/phpshare/classes/entity/_delegate/d_ent_foreign_relations.inc [line:69  calling:query_helper]
fetch_extra_data__impl()/opt/em7/phpshare/classes/entity/_delegate/d_ent_foreign_relations.inc [line:69  calling:query_helper]
_fetch_multi_where()/opt/em7/phpshare/classes/entity/entity_common.inc [line:398  calling:autofetch_all]
_base_fetch_join_fields__impl()/opt/em7/phpshare/classes/entity/entity_common.inc [line:1123  calling:autofetch_all]
_base_fetch_join_fields__impl()/opt/em7/phpshare/classes/entity/entity_common.inc [line:1123  calling:autofetch_all]
fetch_extra_data__impl()/opt/em7/phpshare/classes/entity/_delegate/d_ent_foreign_relations.inc [line:69  calling:query_helper]
fetch_extra_data__impl()/opt/em7/phpshare/classes/entity/_delegate/d_ent_foreign_relations.inc [line:69  calling:query_helper]
fetch_extra_data__impl()/opt/em7/phpshare/classes/entity/_delegate/d_ent_foreign_relations.inc [line:69  calling:query_helper]
_fetch_multi_where()/opt/em7/phpshare/classes/entity/entity_common.inc [line:398  calling:autofetch_all]
_fetch_multi_where()/opt/em7/phpshare/classes/entity/entity_common.inc [line:398  calling:autofetch_all]
_get_extended_tables()/opt/em7/phpshare/classes/entity/_delegate/d_ent_custom_fields.inc [line:288  calling:autofetch_column]
require()/opt/em7/gui/ap/includes/auth.inc [line:53  calling:autofetch_row]
_fetch_multi_where()/opt/em7/phpshare/classes/entity/entity_common.inc [line:398  calling:autofetch_all]
_fetch_multi_where()/opt/em7/phpshare/classes/entity/entity_common.inc [line:398  calling:autofetch_all]
_get_extended_tables()/opt/em7/phpshare/classes/entity/_delegate/d_ent_custom_fields.inc [line:288  calling:autofetch_column]
_fetch()/opt/em7/phpshare/classes/system/deepcache.inc [line:76  calling:autofetch_value]
_fetch()/opt/em7/phpshare/classes/system/deepcache.inc [line:76  calling:autofetch_value]
load()/opt/em7/phpshare/classes/system/system_settings.inc [line:35  calling:autofetch_row]
require()/opt/em7/gui/ap/includes/get_ticket_severity_list.inc [line:5  calling:query]
require_once()/opt/em7/gui/ap/includes/get_ticket_definitions.inc [line:22  calling:query]
fetch_data()/opt/em7/gui/ap/classes/db_registry_table.inc [line:410  calling:autofetch_value]
fetch_data()/opt/em7/gui/ap/classes/db_registry_table.inc [line:427  calling:autofetch_all]
_fetch_multi_where()/opt/em7/phpshare/classes/entity/entity_common.inc [line:398  calling:autofetch_all]
_fetch_multi_where()/opt/em7/phpshare/classes/entity/entity_common.inc [line:398  calling:autofetch_all]
_get_extended_tables()/opt/em7/phpshare/classes/entity/_delegate/d_ent_custom_fields.inc [line:288  calling:autofetch_column]

```

- scanning through run1.txt, we find these two queries which are obviously retrieving ticket template information, and we guess that the php code running those queries (/opt/em7/gui/ap/classes/db_registry_table.inc, line 423) is part of the main php execution path responsible for rendering the ticket templating page:

```
fetch_data()/opt/em7/gui/ap/classes/db_registry_table.inc [line:407  calling:autofetch_value]
SELECT  COUNT(*) FROM master_biz.ticket_templates ts
 LEFT JOIN master_access.accounts a ON (ts.edit_user=a.uid)
 LEFT JOIN master_biz.user_information a_info ON (a_info.uid = a.uid)
 LEFT JOIN master.auth_userstore_attribute_map auth_resource ON (auth_resource.resource_id = a.resource_id)
WHERE 1 LIMIT 1
SIZE: 306 bytes
ROWS:1|ERRNO:0|MSG:
DURATION: 0.00036 seconds
```

```

fetch_data()/opt/em7/gui/ap/classes/db_registry_table.inc [line:423 calling:autofetch_all]
SELECT ts.name, ts.sch_id, ts.ticket_descr, ts.ticket_severity, IF(auth_resource.userdisplayname="disable" OR a_info.userdisplayname IS NULL OR a_info.userdisplayname="", a.user, a_info.userdisplayname) AS user, UNIX_TIMESTAMP(ts.edit_date) AS edit_date, (template_feature = 2 AND (SELECT v.removed FROM scheduler.vevent v WHERE v.removed = 0 AND (v.dtstart > NOW() OR v.recur_until > NOW()) LIMIT 1)) AS sch_pending, (template_feature = 1 AND (SELECT ticket_template FROM master.policies_actions_ticket WHERE ticket_template = ts.sch_id LIMIT 1)) AS automated, ts.ppguid
FROM master_biz.ticket_templates ts
 LEFT JOIN master_access.accounts a ON (ts.edit_user=a.uid)
 LEFT JOIN master_biz.user_information a_info ON (a_info.uid = a.uid)
 LEFT JOIN master.auth_userstore_attribute_map auth_resource ON (auth_resource.resource_id = a.resource_id)
WHERE 1
ORDER BY name ASC LIMIT 50
SIZE: 880 bytes
ROWS:1|ERRNO:0|MSG:
DURATION: 0.00058 seconds

```

We see that line 423 in that file, in our git working tree, is the last line in the following quote:

```
  public function fetch_data($fetch_total = TRUE,$for_report = FALSE) {
    $db = mysqli_helper::obj();

    $GET_array = $this->my_get_array();

    list($forcetables) = $this->get_tables();

    if ($fetch_total) {

      $this->_total_rows = $db->autofetch_value('SELECT '
        .(empty($this->_group_by_fields) ? ' COUNT(*) ' : " COUNT(DISTINCT ".implode(', ',$this->_group_by_fields).') ')
        .$this->_builder->getfromwhere(NULL, $forcetables));
    }

    if (!empty($GET_array['suppress_body'])) {
      $this->add_metadata('bodySuppressed', true);
    } else {
        if($this->is_union)
        {
            $sql = $this->build_union_query();
        }
        else
        {
            $sql = $this->build_query();
        }
      if(!$for_report)
      {
        $this->_rows = $db->autofetch_all($sql);

```

Deploying PHP code from a git working tree into a running appliance

In the next section, we are going to add calls to devlog functions at strategic points in the code (revealed above). We will need to deploy this code into our test stack' CDB so we can exercise it from there.

From inside the G3code directory of a git working tree, assuming we are targeting a CDB at 10.2.4.50, we would run the following rsync:

```
rsync -rtlv -essh --exclude "*.swp" --perms --progress gui phpshare root@10.2.4.50:/opt/em7/
```

Before this will work, you will need to uncomment this line in /etc/ssh/sshd_config

```
PermitRootLogin yes
```

and then bounce the sshd server with:

```
systemctl restart sshd
```

Generating Tracebacks

Now that we know which line of code the main query defining the structure of the ticketing template registry page was called from, we want to understand how that line of code was arrived at. We want to know its execution model.

To answer that, we are going to leverage php's debug_backtrace() function. Here is a simple demo of how that works:

```
#!/usr/bin/php
<?php
// file_summary: debug_backtrace() generates a backtrace

class abc {
   function a() {
      $this -> b();
   }

   function b() {
      $this -> c();
   }

   function c() {
      $x= debug_backtrace();
      echo "backtrace= ".print_r($x, TRUE);
   }
}

(new abc()) -> a();
?>

```

running that program produces this output:

```
(
    [0] => Array
        (
            [file] => /mnt/hd/network_drive/chris/repos/programs/php/test26.php
            [line] => 11
            [function] => c
            [class] => abc
            [object] => abc Object
                (
                )

            [type] => ->
            [args] => Array
                (
                )

        )

    [1] => Array
        (
            [file] => /mnt/hd/network_drive/chris/repos/programs/php/test26.php
            [line] => 7
            [function] => b
            [class] => abc
            [object] => abc Object
                (
                )

            [type] => ->
            [args] => Array
                (
                )

        )

    [2] => Array
        (
            [file] => /mnt/hd/network_drive/chris/repos/programs/php/test26.php
            [line] => 20
            [function] => a
            [class] => abc
            [object] => abc Object
                (
                )

            [type] => ->
            [args] => Array
                (
                )

        )

```

```
)

```

So, debug_backtrace() returns an associative array, with the first element detailing the deepest line of code reached, and every subsequent element walking up the call stack one level. The EM7's devlog class (G3code/phpshare/classes/system/devlog.inc) has a method that will gather, format, and log a backtrace

```
  static public function btrc ($msg=NULL, $stackdepth = 1,$forcelevel=LOG_DEBUG) {
    $msg = (array) $msg;
    foreach(array_reverse(debug_backtrace()) as $call) {
      $args = array();
      foreach ($call['args'] as $arg) {
        $args[] = gettype($arg);
      }
      $msg[] = $call['function'] . '(' . implode(',',$args) . ') at '.$call['file'].':'.$call['line'];
    }
    array_pop($msg); // dont need the btrc() call
    self::_log_syslog(self::$_lvlint[$forcelevel], $msg, $forcelevel, $stackdepth);
  }
```

We write a new method that calls this one:

```
  static public function fixmelog ($msg, $bt= FALSE){
    if ($bt){
      self::btrc($msg, 1, LOG_ERR);
    } else {
      self::err($msg);
    }
  }
```

And call it from the top of the function we just identified by following the mysqli log

```
 public function fetch_data($fetch_total = TRUE,$for_report = FALSE) {
   devlog::fixmelog("db_registry_table.fetch(): enter", TRUE);
   $db = mysqli_helper::obj();
   ...
    devlog::fixmelog("db_registry_table.fetch(): exit");
   return $this;

```

We then

- enable php debug logging again
- tail the devlog:

```
tail -F /data/logs/em7php.log
```

- reload the URL for the page

```
https://10.2.4.50/em7/index.em7?exec=registry&act=registry_ticket_jobs&ap2_embedded=1
```

- resulting in this:

```
19:35:42.44882|/opt/em7/gui/ap/www/em7/index.em7:__main              |42  |info|index.em7 called, request URI:/em7/index.em7?exec=registry&act=registry_ticket_jobs&ap2_embedded=1
19:35:42.47492|*m7/phpshare/classes/system/authuser.inc:auth_by_array|123 |info|auth_by_array using:user,uid
19:35:42.48078|*m7/phpshare/classes/system/authuser.inc:auth_by_array|163 |info|Valid user loaded. Authentication  Passed
15:35:42.48470|*m7/phpshare/classes/system/authuser.inc:auth_by_array|123 |info|auth_by_array using:uid
15:35:42.48486|*m7/phpshare/classes/system/authuser.inc:auth_by_array|163 |info|Valid user loaded. Authentication  Passed
15:35:42.48534|/opt/em7/gui/ap/www/em7/index.em7:__main              |158 |notice|START:index.em7::mem_usage:1.13MiB
15:35:42.48975|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |91  |error|Array
15:35:42.48990|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |91  |error|(
15:35:42.48999|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |91  |error|    [0] => "db_registry_table.fetch(): enter"
15:35:42.49005|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |91  |error|    [1] => "include(string) at /opt/em7/gui/ap/www/em7/index.em7:163"
15:35:42.49010|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |91  |error|    [2] => "require(string) at /opt/em7/gui/ap/includes/liquid_exec_common.inc:38"
15:35:42.49016|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |91  |error|    [3] => "phtml_include(string,array) at /opt/em7/gui/ap/exec/common_regtable.inc:80"
15:35:42.49021|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |91  |error|    [4] => "page_include(string,array) at /opt/em7/gui/ap/includes/functions_core.inc:1364"
15:35:42.49025|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |91  |error|    [5] => "include(string) at /opt/em7/gui/ap/includes/functions_core.inc:1332"
15:35:42.49030|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |91  |error|    [6] => "require(string) at /opt/em7/gui/ap/phtml/common_regtable.phtml:4"
```

```
15:35:42.49035|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |91  |error|    [7] => "echo_init_script() at /opt/em7/gui/ap/phtml/common_regtable_liquid.phtml:39"
15:35:42.49040|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |91  |error|    [8] => "get_dynamic_json() at /opt/em7/gui/ap/classes/registry_table.inc:573"
15:35:42.49045|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |91  |error|    [9] => "get_dynamic_array() at /opt/em7/gui/ap/classes/registry_table.inc:857"
15:35:42.49049|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |91  |error|    [10] => "fetch_data() at /opt/em7/gui/ap/classes/db_registry_table.inc:565"
15:35:42.49054|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |91  |error|    [11] => "fixmelog(string,boolean) at /opt/em7/gui/ap/classes/db_registry_table.inc:399"
15:35:42.49059|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |91  |error|)
15:35:42.49063|*/opt/em7/phpshare/classes/system/devlog.inc:fixmelog |91  |error|

```

- we are now have the information we need (file and line numbers) to place the comments FIXME_INFO_ISSUE_3_10 through 3_19.
- which enables us to,with some manual editing of the output of "fixme_report.sh | fixme_struct.sh 3", arrive at the execution model quoted at the top of this section: PHP Debugging | telling the story (asking and answering questions, FIXME_INFO_ISSUE_X_Y)

Theory

nginx config files and directories

/etc/nginx/nginx.conf

The main nginx config file, this pulls in config files from the following directories

```
    include /etc/nginx/conf.d/*.conf;
    include /opt/em7/share/config/nginx.d/*.conf;

```

/etc/nginx/conf.d

contains

```
em7ngx_pma.conf
em7webconfig.conf
em7ngx_web_ui.conf
em7ngx_em7proxy_web_ui.conf
em7_sladmin.conf
stub_status.passwd
default.conf
em7_restrict_nonsecure.conf
em7_limits.conf
em7_certificate.conf

```

/etc/em7/share/config/nginx.d/

contains

```
em7ngx_web_ui.fragment
em7ngx_em7proxy_web_ui.fragment
em7webconfig.fragment
mimetypes.conf
nextui.fragment
stub_status.fragment
stub_status.passwd
em7_hide_server_tokens.conf
em7ngx_pma.fragment
em7_sladmin.fragment

```

/etc/nginx/conf.d/em7ngx_web_ui.conf

governs the UI, listening on ports 80 and 443 and also pulling in

- `/opt/em7/share/config/nginx.d/em7ngx_web_ui.fragment;`
- `/opt/em7/share/config/nginx.d/stub_status.fragment;`

```
em7ngx_web_ui.conf
server {
    rewrite ^/em7/libs/od_templates/populated/(.+)$ /file_auth_wrapper.em7?redirect=1&match_user=1?  last;

    include  /opt/em7/share/config/nginx.d/em7ngx_web_ui.fragment;
    include  /opt/em7/share/config/nginx.d/stub_status.fragment;

```

```
    listen   80 default_server;
    listen  443 ssl http2 default_server;
    listen   [::]:80 default_server;
    listen  [::]:443 ssl http2 default_server;

    if ($nonsecure) {
        return 403;
    }

    access_log syslog:server=unix:/var/run/rsyslog.sock,facility=local7,tag=em7silo_nginx,severity=info,nohostname;
    error_log syslog:server=unix:/var/run/rsyslog.sock,tag=em7silo_nginx,nohostname info;

    ssl_certificate      /etc/nginx/silossl.pem;
    ssl_certificate_key  /etc/nginx/silossl.key;

    # support for CAC/Client Certificate authentication
    ssl_client_certificate  /var/lib/em7/certs/em7_combined.crt;
    ssl_verify_client       optional_no_ca;
    ssl_verify_depth        2;
    ssl_ciphers HIGH:!aNULL:!MD5:!EXPORT;
    ssl_prefer_server_ciphers on;
    ssl_protocols TLSv1.2 ;
    server_name_in_redirect on;
}

```

/opt/em7/share/config/nginx.d/em7ngx_web_ui.fragment

A lot of the major config details that govern how URLs in http requests are handled/routed internally are found in this file (see the bottom of this section for a full quote of that file).

- a URL that hits the root exactly (“location =/“) is redirected (through the proxy_pass directive) to “http://localhost:3000/default“, which is where a node.js server runs on a CDB
- the most general URL path (“location /") will be directed toward a php file under /usr/local/silo/gui/ap/www, with the default index being index.em7 (the php file found in the v8 repo at “G3code/gui/ap/www/index.em7”
  - if you look at this file on a CDB at path /usr/local/silo/gui/ap/www/index.em7, you won’t see php plaintext. You will see an encrypted php op-code file.
  - if you look at this file in a v8 em7 repo at path G3code/gui/ap/www/index.em7, you will see the plaintext php code that was op-code compiled and then encrypted in the build process.
  - the particular form of op-code caching that em7 uses was developed in-house as a php extension in this repo: https://github.sciencelogic.com/SL1/php-silo
- a typical URL that refers to a php file ending in .em7 is going to be handled by this location block (because it matches the regular expression \.em7, and is more specific than the earlier location block with prefix “/”)

```
location ~ \.em7 {
        fastcgi_split_path_info ^(.+\.em7)(/.+)$;
        fastcgi_pass unix:/var/run/php-fpm/em7.sock;
        fastcgi_index index.em7;
        fastcgi_param SCRIPT_FILENAME /usr/local/silo/gui/ap/www$fastcgi_script_name;
        fastcgi_param SSL_CLIENT_VERIFY $ssl_client_verify;
        fastcgi_param SSL_CLIENT_S_DN $ssl_client_s_dn;
        fastcgi_param SSL_CLIENT_S_DN_CN $ssl_client_username;
        fastcgi_param SSL_CLIENT_CERT $ssl_client_raw_cert;
        fastcgi_read_timeout 120;
        include fastcgi_params;
    }
```

  - this block passes the request from nginx to the php-fpm server via the fastcgi interface

We quote “em7ngx_web_ui.fragment“ file in full below:

```
# Standard EM7 UI nginx config

    client_max_body_size  256M;
    disable_symlinks  off;
    #charset koi8-r;

    #do we want/need SL static 4xx/5xx pages?
    #error_page  404              /404.html;
    #error_page   500 502 503 504  /50x.html;
```

```

    # if they go exactly to / then try to redirect them to the default ui
    location = / {
        # try to navigate to the :3000/default express route
        # that route will redirect back to /ap2 or /em7 depending on what the user
        # has selected in system>settings>behavior
        proxy_set_header Host $host;
        proxy_set_header X-Forwarded-For $remote_addr;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_connect_timeout 10;
        proxy_read_timeout 300;
        proxy_pass http://localhost:3000/default;

        # if we get a 50x error from the express server serving /default,
        # do a 302 redirect to /em7.  if ap2 is set as the default ui, you won't
        # redirect to /ap2 as expected but at least you will see the /em7 ui instead
        error_page 500 502 503 504 @redirect_to_em7;
    }

    # redirect to /em7 - used for when the mechanism for routing to the default ui is not running
    location @redirect_to_em7 {
        # do a 302 redirect to /em7
        return 302 $scheme://$host/em7;
    }

    # if they try to go to anything with / as the prefix (other than exactly /) then try to serve up some file or dir
    location / {
        server_name_in_redirect off;
        root   /usr/local/silo/gui/ap/www;
        index  index.em7;
    }

    location /em7/license.html {
        add_header Cache-Control "private, no-cache, must-revalidate, no-store, max-age=0, post-check=0, pre-check=0";
        add_header Pragma "no-cache";
        add_header Expires "0";
        root       /usr/local/silo/gui/ap/www;
    }

    location ~ ^\/api {
        fastcgi_split_path_info ^(/api)(.+)$;
        fastcgi_pass unix:/var/run/php-fpm/em7.sock;
        fastcgi_param SCRIPT_FILENAME /usr/local/silo/gui/is/www/api_main.em7;
        include fastcgi_params;
        fastcgi_param  SCRIPT_NAME /api;
    }
    location ~ ^\/gmapi {
        fastcgi_split_path_info ^(/gmapi)(.+)$;
        fastcgi_pass unix:/var/run/php-fpm/em7.sock;
        fastcgi_param SCRIPT_FILENAME /usr/local/silo/gui/is/www/api_main.em7;
        include fastcgi_params;
        fastcgi_param  SCRIPT_NAME /gmapi;
    }

    location ~ \.em7 {
        fastcgi_split_path_info ^(.+\.em7)(/.+)$;
        fastcgi_pass unix:/var/run/php-fpm/em7.sock;
        fastcgi_index index.em7;
        fastcgi_param SCRIPT_FILENAME /usr/local/silo/gui/ap/www$fastcgi_script_name;
        fastcgi_param SSL_CLIENT_VERIFY $ssl_client_verify;
        fastcgi_param SSL_CLIENT_S_DN $ssl_client_s_dn;
        fastcgi_param SSL_CLIENT_S_DN_CN $ssl_client_username;
        fastcgi_param SSL_CLIENT_CERT $ssl_client_raw_cert;
        fastcgi_read_timeout 120;
        include fastcgi_params;
    }

    # not a bad idea, even if we've phase apache out completely
    location ~ /\.ht {
        deny  all;
    }

```

```
    # this is an internal forward. Any accessing /php_devlog directly won't match this location,
    # but responses from PHP that have X-Accel-Redirect: /php_devlog/* response headers will. This
    # allows PHP to handle access control, but hand off to nginx for gzip and large file xfer
    location /php_devlog {
      internal;
      alias /var/log/em7;
      gzip on;
    }

    # Include nextui
    include /opt/em7/share/config/nginx.d/nextui.fragment;

    # Include streamer prime, if available.
    include /opt/em7/share/config/nginx.d/ngx_streamer_prime*.fragment;

```

php-fpm config files and directories

Intro

php-fpm stands for “PHP FastCGI Process Manager”. It is controlled by systemd. For example, on a test CDB, we have:

```
 systemctl status php-fpm
● php-fpm.service - The PHP FastCGI Process Manager
   Loaded: loaded (/usr/lib/systemd/system/php-fpm.service; enabled; vendor preset: disabled)
   Active: active (running) since Wed 2023-10-04 16:49:19 UTC; 1 weeks 0 days ago
 Main PID: 1202 (php-fpm)
   Status: "Processes active: 0, idle: 29, Requests: 7910, slow: 0, Traffic: 0req/sec"
    Tasks: 30
   Memory: 96.1M
   CGroup: /system.slice/php-fpm.service
           ├─ 1202 php-fpm: master process (/etc/php-fpm.conf)
           ├─ 1555 php-fpm: pool em7-webconfig
           ├─ 1556 php-fpm: pool em7-webconfig
           ├─ 1557 php-fpm: pool em7-webconfig
           ├─ 1558 php-fpm: pool em7-webconfig
           ├─ 1559 php-fpm: pool em7
           ├─ 1560 php-fpm: pool em7
           ├─ 1561 php-fpm: pool em7
           ├─ 1562 php-fpm: pool em7
           ├─ 1563 php-fpm: pool em7
           ├─ 1564 php-fpm: pool em7
           ├─ 1565 php-fpm: pool em7
           ├─ 1566 php-fpm: pool em7
           ├─ 1567 php-fpm: pool em7
           ├─ 1568 php-fpm: pool em7
           ├─ 1569 php-fpm: pool em7
           ├─ 1570 php-fpm: pool em7
           ├─ 1571 php-fpm: pool em7
           ├─ 1572 php-fpm: pool em7
           ├─ 1573 php-fpm: pool em7
           ├─ 1574 php-fpm: pool em7
           ├─ 1575 php-fpm: pool em7
           ├─ 1576 php-fpm: pool em7
           ├─ 1577 php-fpm: pool em7
           ├─ 1578 php-fpm: pool em7
           ├─14201 php-fpm: pool em7
           ├─14222 php-fpm: pool em7
           ├─14223 php-fpm: pool em7
           ├─14261 php-fpm: pool em7
           └─22396 php-fpm: pool em7

```

nginx passes matching http requests to it. This location block in the nginx configs, for example, hands off to php-fpm:

```
location ~ \.em7 {
        fastcgi_split_path_info ^(.+\.em7)(/.+)$;
        fastcgi_pass unix:/var/run/php-fpm/em7.sock;
        fastcgi_index index.em7;
        fastcgi_param SCRIPT_FILENAME /usr/local/silo/gui/ap/www$fastcgi_script_name;
        fastcgi_param SSL_CLIENT_VERIFY $ssl_client_verify;
        fastcgi_param SSL_CLIENT_S_DN $ssl_client_s_dn;
        fastcgi_param SSL_CLIENT_S_DN_CN $ssl_client_username;
        fastcgi_param SSL_CLIENT_CERT $ssl_client_raw_cert;
```

```
        fastcgi_read_timeout 120;
        include fastcgi_params;
    }
```

The main php-fpm process is listening on a unix domain socket for incoming requests:

```
fastcgi_pass unix:/var/run/php-fpm/em7.sock;
```

/etc/php-fpm.conf

This is the primary config file, and at the top we see the pid file and error log locations declared:

```
;;;;;;;;;;;;;;;;;;;;;
; FPM Configuration ;
;;;;;;;;;;;;;;;;;;;;;

; All relative paths in this configuration file are relative to PHP's install
; prefix.

; Include one or more files. If glob(3) exists, it is used to include a bunch of
; files from a glob(3) pattern. This directive can be used everywhere in the
; file.
include=/etc/php-fpm.d/*.conf

;;;;;;;;;;;;;;;;;;
; Global Options ;
;;;;;;;;;;;;;;;;;;

[global]
; Pid file
; Default Value: none
pid = /run/php-fpm/php-fpm.pid

; Error log file
; If it's set to "syslog", log is sent to syslogd instead of being written
; in a local file.
; Default Value: /var/log/php-fpm.log
error_log = /var/log/php-fpm/error.log

...
```

we also see that that other config files residing in /etc/php-fpm.d/ are pulled in.

/etc/php-fpm.d/

There are three files found here:

```
www.conf
em7.conf
em7-webconfig.conf

```

www.conf

This file is the canonical config file that php-fpm ships with where most of the possible options are documented and also commented out. As part of em7's initialization procedure a ";" is prefixed to every line of the file, ensuring that none of the default settings are in effect.

We quote a few lines from the top of this file

```
;## Configuration file disabled by EM7 initialization
;; Start a new pool named 'www'.
;; the variable $pool can we used in any directive and will be replaced by the
;; pool name ('www' here)
;[www]
...
;; The access log file
;; Default: not set
;;access.log = log/$pool.access.log
;
...
```

as it shows how to define the php-fpm access log. This is always disabled in production, and even php-fpm disables it by default, but when experimenting on a test stack and debugging issues where php refuses to process a request, enabling this could shed light on the mechanism involved in a specific bug.

em7.conf

Here we see the error log location being defined (/var/log/em7/php-error.log), and the parameters controlling how many php-fpm processes are started.

```
[em7]

include=/opt/em7/share/config/fpm.em7

pm = dynamic

pm.max_children = 256
pm.start_servers = 20
pm.min_spare_servers = 20
pm.max_spare_servers = 50

pm.max_requests = 2500

php_admin_value[error_log] = /var/log/em7/php-error.log
php_admin_flag[log_errors] = on
php_admin_flag[fastcgi.logging] = off

```

em7-webconfig.conf

As above, we see the error log location being defined (/var/log/em7/webconfig-php-error.log)

```
[em7-webconfig]

include=/opt/em7/share/config/fpm.em7-webconfig

pm = dynamic

pm.max_children = 50
pm.start_servers = 4
pm.min_spare_servers = 4
pm.max_spare_servers = 6

pm.max_requests = 2500

php_admin_value[error_log] = /var/log/em7/webconfig-php-error.log
php_admin_flag[log_errors] = on
php_admin_flag[fastcgi.logging] = off

```

/opt/em7/share/config/fpm.em7

This file was included by /etc/php-fpm.d/em7.conf

```
listen = /run/php-fpm/em7.sock

listen.owner = s-em7-http
listen.group = nginx
listen.mode = 0660

user = s-em7-http
group = s-em7-core

security.limit_extensions = .php .em7

php_admin_value[auto_prepend_file] =
/opt/em7/phpshare/includes/auto_prepend.inc

php_value[output_handler] = ob_gzhandler
php_flag[short_open_tag] = on
php_flag[expose_php] = off
php_value[post_max_size] = 256M
php_value[upload_max_filesize] = 256M
php_value[apc.serializer] = igbinary
php_value[max_execution_time] = 2400
php_value[max_input_vars] = 1250

```

Notes

- the setting “security.limit_extensions = .php .em7” means that only files ending in .php or .em7 will be evaluated by php-fpm. Any other requests making it to this point woudl be rejected.

node.js

Introduction

In the nginx location block quoted above:

```
    location = / {
        # try to navigate to the :3000/default express route
        # that route will redirect back to /ap2 or /em7 depending on what the user
        # has selected in system>settings>behavior
        proxy_set_header Host $host;
        proxy_set_header X-Forwarded-For $remote_addr;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_connect_timeout 10;
        proxy_read_timeout 300;
        proxy_pass http://localhost:3000/default;

        # if we get a 50x error from the express server serving /default,
        # do a 302 redirect to /em7.  if ap2 is set as the default ui, you won't
        # redirect to /ap2 as expected but at least you will see the /em7 ui instead
        error_page 500 502 503 504 @redirect_to_em7;
    }
```

URLs that access http://<CDB ip>/ exactly are redirected to:

```
proxy_pass http://localhost:3000/default;
```

This is a node.js server.

main command line

The command line running the node.js server is:

```
/opt/em7/nextui/node_modules/node/bin/node /opt/em7/nextui/index.js
```

That index.js file contains:

```
import '@sciencelogic/sl-em7-gql/src/server.js';

```

main config file: /opt/em7/nextui/nextui.env

This appears to be the main config file for the node.js server. We quote its contents here:

```
######################################################################
# This environment file WILL BE OVERWRITTEN by every install and upgrade.
#
# Customers SHOULD NOT configure their system using this file. This file
# is for internal use only. Instead, change or add values in the nextui.conf
# file. If the same key exists in both nextui.env and nextui.conf, the value
# in nextui.conf will win.
######################################################################

API_PROXY_HOST=https://localhost
GM_EM7_ENV_ID=RU03MTAxMA==
AUTH_CACHE=300000
AUTH_MODE=local
AUTHENTICATE_PHP_FCGI_SOCK_FILE=/var/run/php-fpm/em7.sock
CONTENT_PACKAGE_LOCATION=/var/opt/em7/gui/nextui
CONTENT_IMPORT_LOCATION=/usr/local/silo/nextui/node_modules/@sciencelogic/ap2/public/content-packages/
PUBLIC_FOLDER=/usr/local/silo/nextui/node_modules/@sciencelogic/ap2/public
GQL_RESOLVE_FEATURE_TOOLS=local
GQL_RESOLVE_USER_PREFERENCES=local
NEXT_UI_LOG_LEVEL=all:warn
NODE_ENV=production
NODE_TLS_REJECT_UNAUTHORIZED=0
PATH=/opt/em7/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/root/bin
PORT=3000
```

```
SELF_UPDATE=enabled
SOCK_FILE_LOCATION=/var/run/em7/nextui-key.sock
SETTINGS_BUSINESS_SERVICES_THRESHOLDS=enabled
GQL_RESOLVE_EVENT_CATEGORIES=enabled
GQL_RESOLVE_DEVICE_RUN_DYNAMIC_APPLICATION=enabled
CREDENTIAL_SUBTYPE_AVAILABLE=enabled

```

main content directory: /usr/local/silo/nextui/node_modules/@sciencelogic

has these contents:

```
drwxr-xr-x.   4 root root  4096 Sep  1 19:31 ap2
drwxr-xr-x.   5 root root  4096 Sep  1 19:31 auth
drwxr-xr-x.   2 root root    63 Sep  1 19:31 em7crypt
drwxr-xr-x.   4 root root    73 Sep  1 19:31 revis-network
drwxr-xr-x.   6 root root  4096 Sep  1 19:31 sl-em7-gql

```

/opt/em7/nextui/index.js

This is the javascript file run by the node.js server:

```
export { middlewares } from './middleware/index.js';
export { primeFeatureToggles, getToggle } from
'./lib/featureToggles/index.js';
export { default as start } from './server.js';

```

/opt/em7/nextui/server.js

```
#!/usr/bin/env node

import cluster from 'cluster';
import path from 'path';
import promClient from 'prom-client';
import log from './log.js';
import features from './features/index.js';
import { workerExited } from './messenger/index.js';
import { primeFeatureToggles } from './lib/featureToggles/index.js';
import { addListener, addListeners, setup, terminateListener } from
'./orchestration.js';
import { waitForHaDrActiveStatus } from
'./waitForHaDrActiveStatus.js';
import dirname from './dirname.js';
import './load-dotenv.js';
import { setUpSharedFailedServices } from
'./lib/businessServices/onDemandProcessing.js';

const __dirname = dirname(import.meta.url); // eslint-disable-line no-
underscore-dangle

promClient.collectDefaultMetrics();

const processesGauge = new promClient.Gauge({
  name: `process_count`,
  help: `count of processes`,
});
processesGauge.set(1);

let started = false;
async function start() {
  if (started) {
    return;
  }
  started = true;
  process.chdir(path.resolve(__dirname, '..'));

  await primeFeatureToggles();

  // restart workers that crash
  cluster.on('exit', async (worker, code) => {
    workerExited(worker.id);
    processesGauge.dec();
    if (code !== 0 && !worker.exitedAfterDisconnect) {
      if (await terminateListener(worker)) {
        log.info(`Worker ${worker.id} crashed. Starting a new
worker`);
        addListener();
      }
    }
  });
```

```

  // Sit here quietly while waiting to become the Active node in a HA/DR setup
  await waitForHaDrActiveStatus();

  await setup();

  const numWorkers = features.toggles.mockServer() ? 1 : features.toggles.numberOfWorkers();

  // create workers
  log.info(`Using a node cluster of ${numWorkers} workers`);

  setUpSharedFailedServices();
  addListeners();
}

setTimeout(start, 10); // auto start if not started elsewhere
export default start;

```

post mortem articles

2024-07-15 EM-60397: Potential Defect - nginx AP proxy_pass to node (IPv6)

Link: 💔EM-60397: Potential Defect - nginx AP proxy_pass to node (IPv6) CLOSED

From the description:

```
Observed Behavior: Nginx errors in /var/log/em7/ngx.log

Expected Behavior: No Nginx errors

Summary of troubleshooting steps taken so far and the current status:  Customer's concern, please see attatched provided logging and config:
Hello,
I am going to go over on the AP node the proxy pass configuration inside the following files :
em7ngx_web_ui.fragment
nextui.fragment
In specific the use of proxy_pass http://localhost:3000 this is where it passes the data over to node due to the fact that localhost is set inside /etc/hosts in 2 locations it may try ipv6 and or ipv4

Adding Additional post to show logs and /etc/hosts file
Expected Outcome :
If node was listening to both ipv6 and ipv4 there would be no errors in the /var/log/ngx.log but since node is listening on only ipv4 it randomly throws errors when it tries the ipv6 connection

Solution
 - edit the proxy pass to do 127.0.0.1 only
 - configure node to bind to both ipv4 and ipv6 network

This is to avoid errors in the log file as I was looking for something else and found this. I do not know what it affects when it tries the ipv6 and what fails during this but can just tell you
from the standpoint of the error log and the random spam it does.
```

2024-03-15 EM-63784: 504 Gateway Timeouts on CDB appliance UI and API calls

Link: EM-63784: 504 Gateway Timeouts on CDB appliance UI and API calls CLOSED

From the description:

```
Observed Behavior: Customer is reporting 504 gateway timeouts when attempting to use the UI or run API calls. Customer has also
noted a high number of sleep queries on the database that return after being killed.
Expected Behavior: UI to be accessible and API calls not fail due to gateway timeouts.
Summary of troubleshooting steps taken so far and the current status:
Initial review: Identified the main issue accessing the UI was getting 504 gateway timeouts when trying to access it.
It worked sometimes, but had the 504 often
We found that there were long sleep queries coming from the AP that were slowing down SQL, leading to 504 gateway timeouts.
These queries were killed, customer insists that there have been no new dashboards or reports configured since the issue began.
longer running query identified:
SELECT ELT(sev, 'Notice', 'Minor', 'Major', 'Critical') as severity, sev, countEmoji :yellow_star::yellow_star: as count
FROM master_logs.system_messages WHERE date_edit > DATE_SUB(NOW(), INTERVAL 1 DAY) GROUP BY sev DESC ORDER BY sev DESC;
checking the count of the master_logs.system_messages returend a value of 13097632
```

2019-12-12 EM-34351: Unable to get DEVLOG from AP

Link: 💔EM-34351: Unable to get DEVLOG from AP CLOSED

From the description:

```
We are trying to troubleshoot login issues with Azure, Okta, and SL1.
Following https://support.sciencelogic.com/s/article/1049 does not seem to permit us to log debug level using the GUI settings on
the admin portal. It only appears to work on the database gui.

If we try to set it manually follow the same instructions / restart nginx, php-fpm we can not get the logs to show up for debug level.

if we delete the lines in silo.conf, restart all services or reboot and then attempt to use ./system_status.sh -d the dash d no
longer exists
```

From this comment: 💔EM-34351: Unable to get DEVLOG from AP CLOSED

```
It really does sound like the UI they used was not hitting the same AP as the one to which Rob SSH'd.  The way to confirm this is
to send a bogus request from the browser (i.e. https://targetsystem/GREPFORTHIS).  you'll get a 404 back and you can look
for "GREPFORTHIS" in the access logs of the system to which you are SSHd

We can set up a webex if you need help finally tracking this one down.  Also, I have a quick and dirty script for turning on devlog
from the CLI instead of the UI (see php_devlog_config.php attached to the  ticket)
```

From this comment: 💔EM-34351: Unable to get DEVLOG from AP CLOSED

```
Customer tested the php_devlog_config.php in their test envitonment and agrees that it meets their needs.  I'd like for this ticket to
be moved into being a support defect instead of an escalation and we'll get the php_devlog_config.php script into the product as a
 supported utility rather than a by-request-only script that support has to hand out to customers.
    it needs a change that forces it to be run as the www user instead of em7admin so that the UI and CLI options for enabling logging
    don't clash
    the -M option needs to be removed from the usage() function OR implemented. Right now, it's advertised as being supported but does
    nothing
Otherwise, the script is working great.
```